Emergency Training Simulator POC

Markdown documentation PDF copy

Minimal Client-Side State Machine

Source: unreal/contract/client-state-machine.md

Markdown stays editable. This PDF is the reading/share copy.

Boundary: this project is still a POC, not operational guidance.

Minimal Client-Side State Machine

This state machine is only for the future Unreal client shell.

Its job is to keep message order clean. It does not implement scenario logic.

```
DISCONNECTED
-> CONNECTING
-> WAITING_FOR_SESSION_STARTED
-> READY_FOR_ACTION
-> WAITING_FOR_STATE_UPDATE
-> READY_FOR_ACTION
-> WAITING_FOR_STATE_UPDATE
-> WAITING_FOR_SCENARIO_END
-> ENDED
```

Any backend error message moves the client to ERROR.

States

| State | Meaning | Allowed client action | | --- | --- | --- | | DISCONNECTED | No socket is open. | Open a WebSocket. | | CONNECTING | The socket is opening. | Wait. | | WAITING_FOR_SESSION_STARTED | start_session has been sent. | Wait for session_started. | | READY_FOR_ACTION | The backend has sent the current state. | Send one trainee_action. | | WAITING_FOR_STATE_UPDATE | One action is in flight. | Wait for scenario_state_update. | | WAITING_FOR_SCENARIO_END | The backend reported terminal state. | Wait for scenario_ended. | | ENDED | Session is complete. | Close or return to menu. | | ERROR | Backend rejected the session/message. | Stop sending actions; open a new session if needed. |

Important boundary

The client can decide what UI to show, but it must not invent scenario progress. scenario_state_update and scenario_ended are the only source of scenario progress.